

Sports Hall Booking System using ServiceNow

Project Report

Submitted By

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Abstract

The Sports Hall Booking System is a ServiceNow-based application developed to simplify the process of booking sports halls. It allows users to reserve halls, manages booking information, prevents duplicate bookings using business rules, and provides reports and dashboards for monitoring bookings. The system reduces manual work, improves booking accuracy, and provides a centralized platform for managing sports hall reservations.

Objectives

- Develop an online sports hall booking application.
- Prevent duplicate bookings.
- Automatically manage booking status.
- Generate reports and dashboards.
- Improve booking management using ServiceNow.

Software Requirements

Component	Requirement
Platform	ServiceNow Personal Developer Instance
IDE	ServiceNow Studio
Browser	Google Chrome
Version Control	GitHub

Modules

1. Custom Table

Table Name

u_sports_hall_bookings

Stores booking information.

Column label	Type	Reference	Max length	Default value	Display
Mode of Payment	String	(empty)	40		false
Reason for Cancellation	String	(empty)	40		false
Created by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Hall Name	String	(empty)	40		false
Refund Payment	Currency	(empty)	20		false
Duration Hours	Decimal	(empty)	20		false
Reminder	True/False	(empty)	40		false
Transaction Number	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Status	Choice	(empty)	40		false
End Time	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false
Mode of Refund	String	(empty)	40		false
Purpose	String	(empty)	40		false
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Start Time	Date/Time	(empty)	40		false
Payment	Currency	(empty)	20		false
Refund Status	String	(empty)	40		false
Insert a new row...					

2. Fields

The custom table contains fields such as:

- Hall Name
- Booked By
- Start Time
- End Time
- Status
- Payment
- Duration Hours
- Purpose
- Mode of Payment
- Refund Status
- Reminder

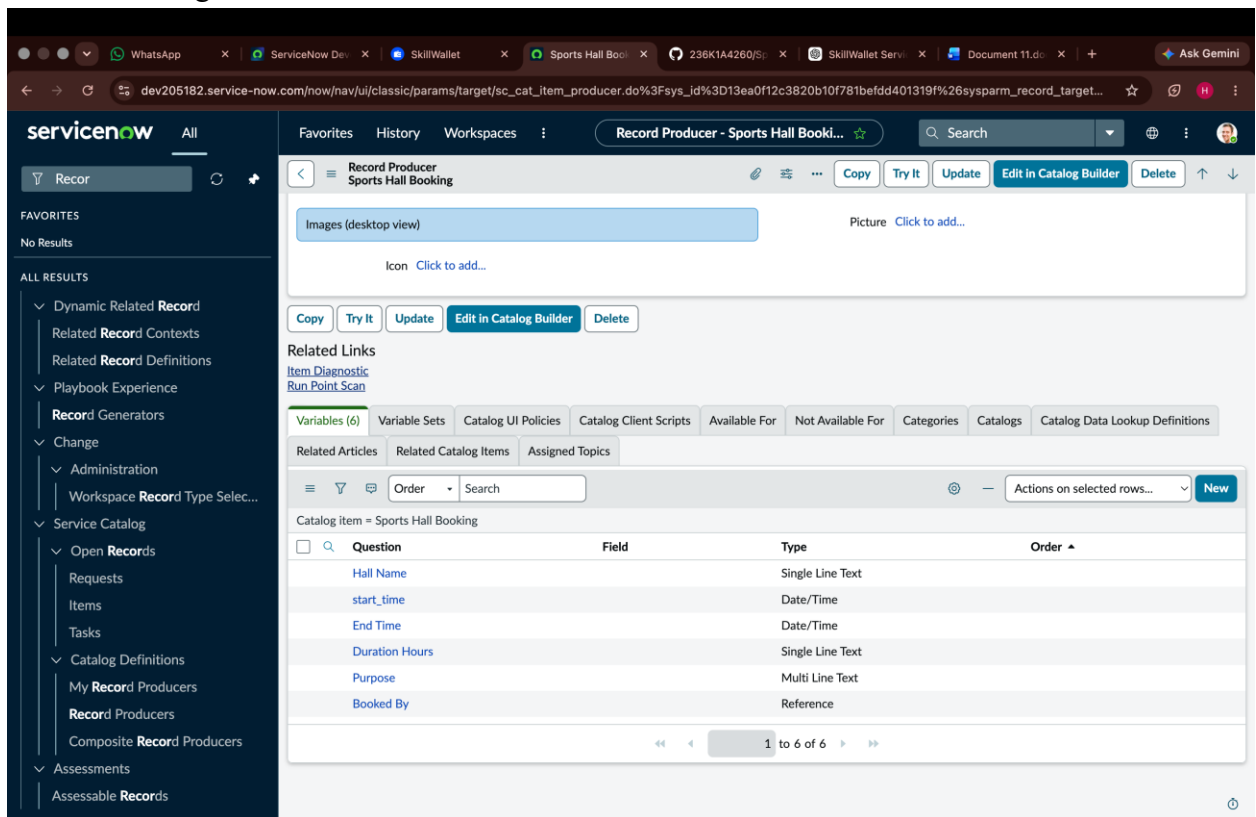
- Reason for Cancellation

The screenshot shows the ServiceNow interface for the 'Table - Sports Hall Bookings'. The left sidebar contains a navigation menu with categories like 'Platform Analytics Administration', 'Data Sources', 'Protected Tables', 'Log Protection', 'Protected Table Log', 'System Archiving', 'Archive Tables', 'System Clone', 'Clone Definition', 'System Definition', 'Tables', 'Table Rotations', 'Decision Tables', and 'Remote Tables'. The main content area displays a table of dictionary entries for the 'Sports Hall Bookings' table. The table has columns for 'Column label', 'Type', 'Reference', 'Max length', 'Default value', and 'Display'. The 'Reason for Cancellation' column is highlighted in blue.

Column label	Type	Reference	Max length	Default value	Display
Mode of Payment	String	(empty)	40		false
Reason for Cancellation	String	(empty)	40		false
Created by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Hall Name	String	(empty)	40		false
Refund Payment	Currency	(empty)	20		false
Duration Hours	Decimal	(empty)	20		false
Reminder	True/False	(empty)	40		false
Transaction Number	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Status	Choice	(empty)	40		false
End Time	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false
Mode of Refund	String	(empty)	40		false
Purpose	String	(empty)	40		false
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Start Time	Date/Time	(empty)	40		false
Payment	Currency	(empty)	20		false
Refund Status	String	(empty)	40		false

3. Record Producer

A Record Producer is used to allow users to submit booking requests easily through the Service Catalog.

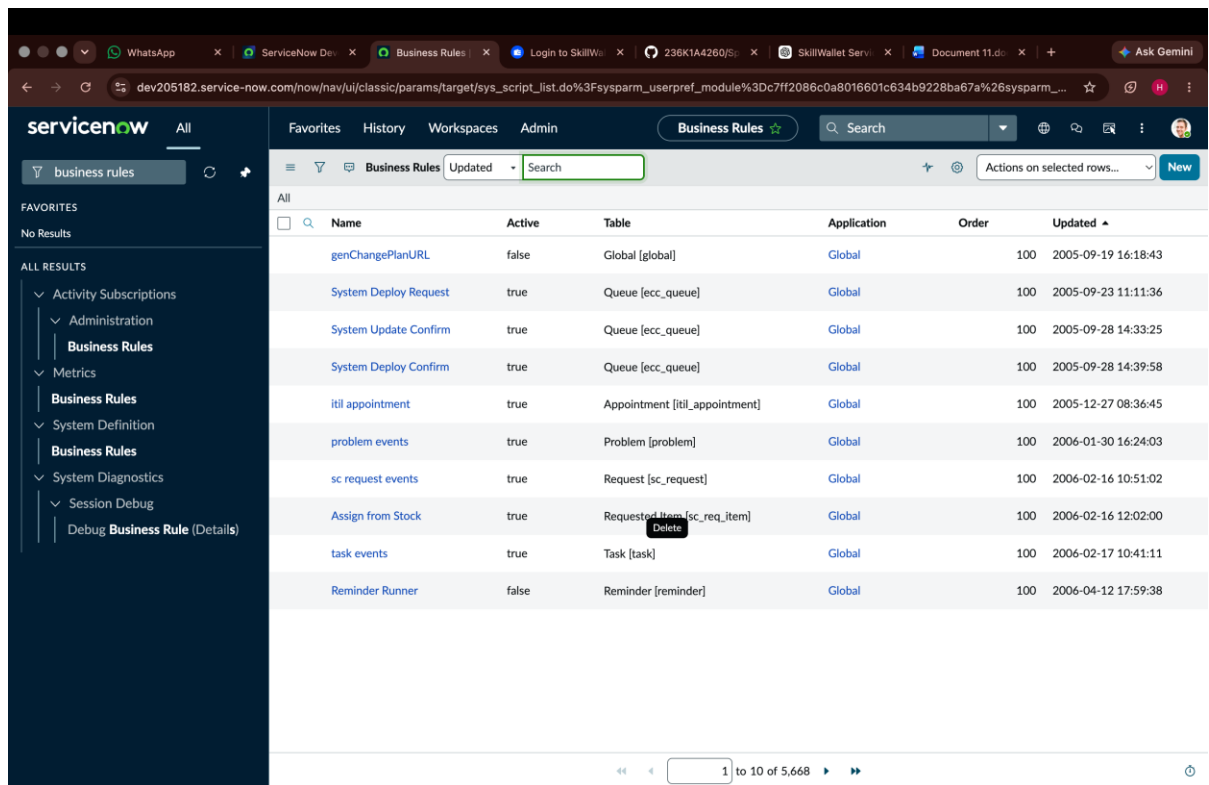


4. Business Rules

Three Business Rules were implemented:

- Prevent Duplicate Booking
- Auto Status Update

- Auto Fee Calculation



The screenshot shows the ServiceNow Business Rules interface. The left sidebar contains a navigation menu with 'business rules' selected. The main area displays a table of business rules. The table has columns for Name, Active, Table, Application, Order, and Updated. The rules listed include 'genChangePlanURL', 'System Deploy Request', 'System Update Confirm', 'System Deploy Confirm', 'itil appointment', 'problem events', 'sc request events', 'Assign from Stock', 'task events', and 'Reminder Runner'.

Name	Active	Table	Application	Order	Updated
genChangePlanURL	false	Global [global]	Global	100	2005-09-19 16:18:43
System Deploy Request	true	Queue [ecc_queue]	Global	100	2005-09-23 11:11:36
System Update Confirm	true	Queue [ecc_queue]	Global	100	2005-09-28 14:33:25
System Deploy Confirm	true	Queue [ecc_queue]	Global	100	2005-09-28 14:39:58
itil appointment	true	Appointment [itil_appointment]	Global	100	2005-12-27 08:36:45
problem events	true	Problem [problem]	Global	100	2006-01-30 16:24:03
sc request events	true	Request [sc_request]	Global	100	2006-02-16 10:51:02
Assign from Stock	true	Request Item [sc_req_item]	Global	100	2006-02-16 12:02:00
task events	true	Task [task]	Global	100	2006-02-17 10:41:11
Reminder Runner	false	Reminder [reminder]	Global	100	2006-04-12 17:59:38

5. Service Portal

A Service Portal was created containing pages for:

- Home
- Book Hall
- My Bookings

- Booking Details

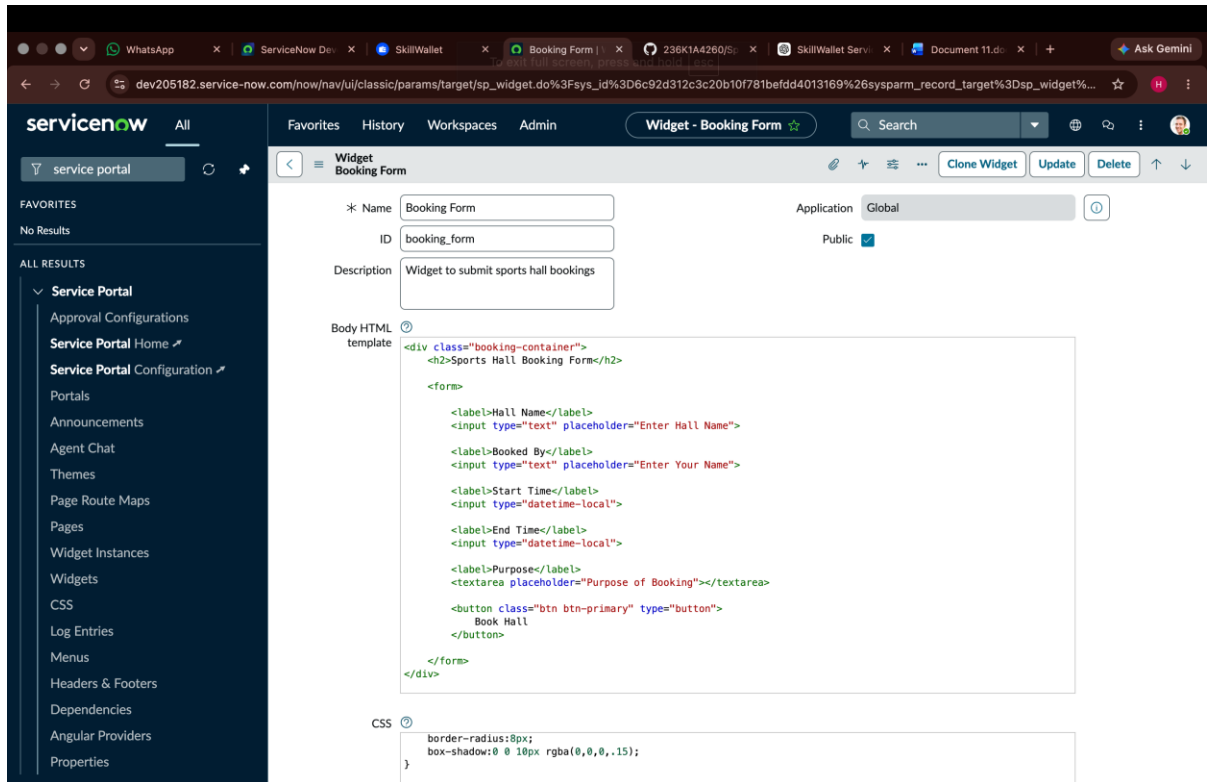
The screenshot displays the ServiceNow configuration interface for a 'Sports Hall Booking Portal'. The left sidebar shows the 'Service Portal' menu with options like 'Approval Configurations', 'Service Portal Home', 'Service Portal Configuration', 'Portals', 'Announcements', 'Agent Chat', 'Themes', 'Page Route Maps', 'Pages', 'Widget Instances', 'Widgets', 'CSS', 'Log Entries', 'Menus', 'Headers & Footers', 'Dependencies', 'Angular Providers', and 'Properties'. The main configuration area includes fields for 'Title' (Sports Hall Booking Portal), 'URL suffix' (sports), 'Homepage' (book_hall), 'Application' (Global), 'KB home page', '404 page', 'Login page', 'Catalog home page', 'Logo' (Click to add...), 'Catalog category home page', 'Logo Alt Text' (Sports Hall Booking Portal), 'Main menu', 'Icon' (Click to add...), 'Theme' (La Jolla), 'Inactive' checkbox, 'Quick start config' (with minus and plus buttons), 'CSS variables' (with a plus icon), 'Hide portal name' checkbox, and 'Support right to left' checkbox.

6. Widgets

Custom widgets were created:

- Booking Form
- Available Halls
- My Bookings

- Calendar



7. Notifications

Email notifications were configured for:

- Booking Confirmation
- Booking Cancellation

8. Reports

Reports developed include:

- Total Bookings
- Booking Status

- Hall-wise Bookings

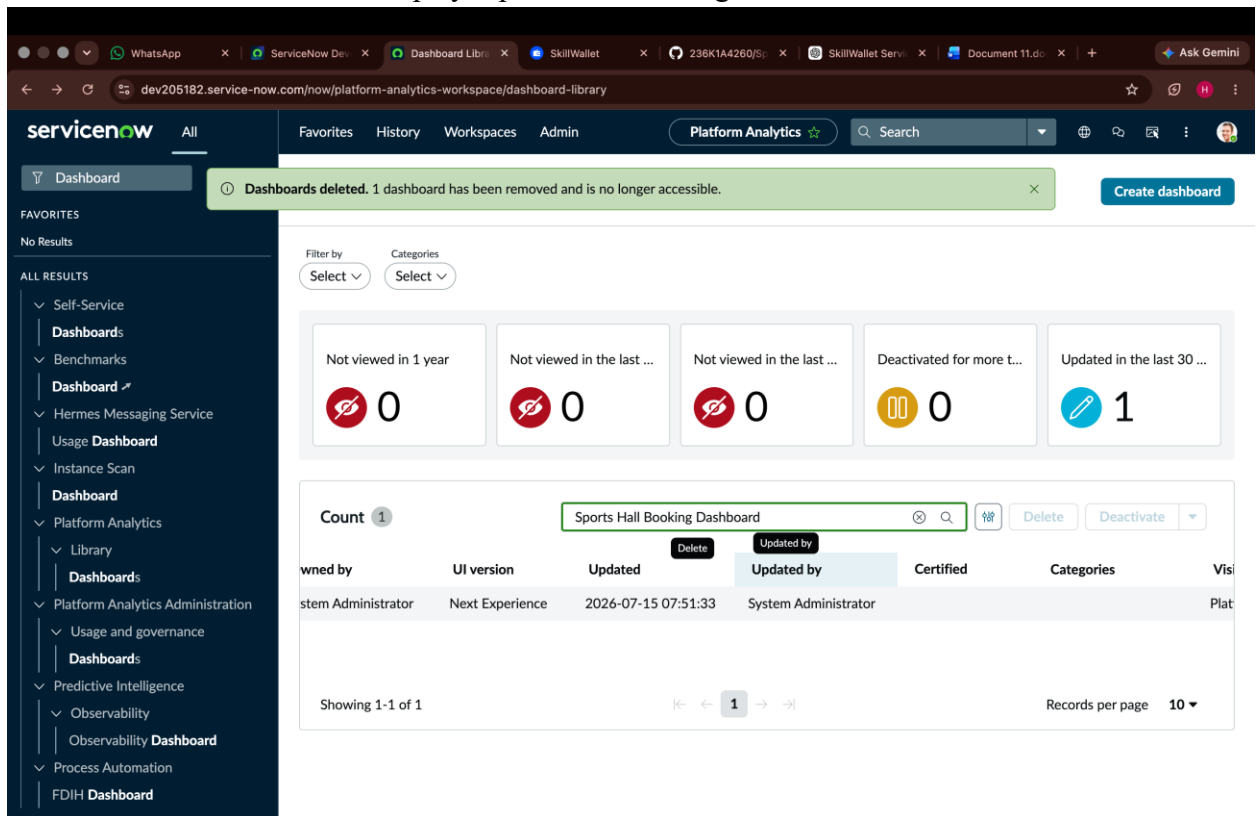
The screenshot displays the ServiceNow interface for configuring a report titled "Booking Status". The left sidebar contains a navigation menu with various options including "Self-Service", "Business Applications", "Dashboards", "Service Catalog", "Employee Center", "Knowledge", "Visual Task Boards", "Incidents", "Watched Incidents", "My Requests", "Requested Items", "Watched Requested Items", "My Connected Apps", "My Profile", "My Tagged Documents", "My Tags", "My Knowledge Articles", "Take Survey", "My Approvals", "My Assessments & Surveys", and "My Assets Analytics". The main content area shows the report configuration details:

- Application:** Global
- Title:** Booking Status
- Table:** Sports Hall Bookings [u_sports_hall_bookings]
- Field Name:** u_status
- Type:** Donut
- Chart size:** Large
- User:** 6816f79cc0a8016401c5a33be04be441
- Filter:** Includes buttons for "Add Filter Condition" and "Add OR Clause". Below these are dropdowns for "-- choose field --", "-- oper --", and "-- value --".
- Roles:** Includes an edit icon.

At the bottom of the configuration area, there are buttons for "Update", "View Report", "Edit Report", and "Delete".

9. Dashboard

A dashboard was created to display reports and booking statistics.



Business Rules Implemented

Prevent Duplicate Booking

Checks whether another booking exists for the same hall and time slot.

Auto Status Update

Automatically sets the booking status to **Booked** when a new booking is created.

Auto Fee Calculation

- Reduces manual booking process.
- Prevents duplicate reservations.
- Easy booking management.
- Automated notifications.
- Dashboard for monitoring bookings.
- Generates useful reports.

Future Enhancements

- Online payment gateway integration.
- QR code-based booking verification.
- Mobile application.
- SMS notifications.
- Calendar synchronization.

Conclusion

The Sports Hall Booking System successfully automates the sports hall reservation process using ServiceNow. The application improves efficiency by reducing manual work, preventing duplicate bookings, and providing real-time reports and dashboards. The project demonstrates practical implementation of ServiceNow components such as custom tables, business rules, record producers, notifications, reports, dashboards, and the Service Portal.

References

1. ServiceNow Documentation
2. ServiceNow Developer Portal
3. GitHub
4. ServiceNow Community